

InPlace AI: Technical Requirements

Note: Numeric thresholds (e.g., limits, timeouts, retention defaults) are **initial targets** and may be adjusted based on LandSeed operational preferences.

Functional Requirements

1. Client-Facing Digital Intake System

FR-1.1: The system shall provide a web-based intake form that requires clients (seniors and caregivers) to create an account before submitting a modification request. Account creation shall be streamlined with minimal required fields (name, email, phone number) to reduce friction.

FR-1.2: Clients shall be able to upload multiple photos (minimum 1, maximum 10) of the space requiring modification, with each photo not exceeding 10MB. Accepted file types shall be limited to JPG, JPEG, PNG, and HEIC formats.

FR-1.3: The intake form shall accept optional floor plans and layout sketches in common formats (PDF, JPG, PNG), with a maximum file size of 15MB per document.

FR-1.4: Clients shall be able to describe their modification needs through guided questions rather than open-ended text fields.

FR-1.5: The system shall provide real-time validation and helpful error messages for incomplete or invalid submissions.

FR-1.6: Clients shall be able to save their intake form as a draft and resume completion later without losing entered information.

FR-1.7: Clients shall be able to delete or replace uploaded photos before final submission.

FR-1.8: The system shall allow clients to cancel a draft submission and clear all entered data.

FR-1.9: When a caregiver is submitting on behalf of a senior, the intake form shall require the caregiver to identify the senior (name, relationship) and confirm their authority and consent to submit the request on the senior's behalf.

FR-1.10: The intake form shall require client consent to collect, store, and process photos and personal information, with a plain-language summary of how the data will be used.

FR-1.11: All uploaded files shall be scanned for viruses and malware before being stored in the system. Files that fail security scanning shall be rejected with a clear error message to the client.

FR-1.12: The intake form shall require clients to provide the service address (street address, city, province, postal code) where the modification work will be performed.

FR-1.13: The intake form shall require clients to specify their property ownership status (owner, tenant, other). For tenant-occupied properties, the system shall prompt for landlord contact information.

2. AI-Driven Estimation & Pricing

FR-2.1: The system shall support modification requests containing multiple modification items or areas (e.g., grab bars + raised toilet + walk-in shower). The AI may propose multiple items based on photo analysis, and staff shall be able to add or remove items. Pricing shall be itemized per modification item.

FR-2.2: The system shall analyze uploaded photos to identify the type and scope of modifications needed (grab bars, ramps, roll-in showers, etc.).

FR-2.3: The pricing engine shall generate quotes based on the predefined pricing matrix, including material costs, labor costs, and appropriate markup tiers.

FR-2.4: The system shall provide tiered pricing options (economy, standard, premium) where applicable.

FR-2.5: Estimate delivery shall occur in three stages:

- **FR-2.5a:** After the client finalizes intake, the system shall generate an initial automatic estimate immediately, displayed as a price range (min-max) rather than a single fixed price.
- **FR-2.5b:** The system shall generate a second, more refined estimate within 4-24 hours during business hours (9:00 AM-5:00 PM EST, Monday-Friday), or by the next business day for after-hours submissions.
- **FR-2.5c:** The system shall require a consultation step before producing a finalized project budget.

FR-2.6: The system shall flag projects for manual review by the Client Advisory Team when: (a) the project complexity exceeds automation thresholds, or (b) the AI analysis has low confidence in identifying the modification type or scope.

FR-2.7: Each generated quote shall record and store the pricing matrix version and grant rules version used at the time of generation. All changes to pricing matrices and grant rules shall be auditable, recording who made the change, when it was made, and what was changed.

FR-2.8: Staff shall be able to complete a modification request end-to-end in manual mode, including the ability to attach custom drawings, manually edit project scope and pricing, and generate the complete output package (quote + grant documents or placeholder + work order package) without relying on AI analysis.

3. Grant Eligibility & Paperwork Automation

FR-3.1: The system shall automatically assess project eligibility against CMHC and provincial grant criteria based on modification type and client information.

FR-3.2: For eligible projects, the system shall generate initial grant application paperwork with pre-filled client and project details.

FR-3.3: The grant module shall provide clear explanations of which grants apply and estimated funding amounts.

FR-3.4: Clients shall be able to download generated grant documents in PDF format through secure, time-limited download links.

FR-3.5: The system shall support grant application lifecycle tracking by maintaining a configurable set of statuses (e.g., Draft, Submitted, Awaiting Client Info, Awaiting Third Party, Approved, Rejected), recording timestamps and the last-updated-by user, and notifying clients via email when their application requires action or when a status changes.

FR-3.6: The system shall allow clients or staff to upload required supporting documents (proof of income, medical documentation, property ownership, etc.) for grant applications, with each document tracked and associated with the specific grant application.

FR-3.7: The system shall provide a mechanism for clients or staff to capture required signatures and acknowledgements for grant submission, maintaining a timestamped record of who provided each signature and when.

4. Admin Panel & Manual Override

FR-4.1: The system shall support role-based access control with distinct permissions for at minimum: Client, Client Advisory Team, Internal Grant Team, Project Manager/Operations, and System Administrator roles.

FR-4.2: Admin users shall have access to a dashboard displaying all pending, in-progress, and completed project requests.

FR-4.3: The Client Advisory Team shall be able to review AI-generated estimates and manually adjust pricing, scope, or grant eligibility.

FR-4.4: Admins shall be able to add internal notes and comments visible only to LandSeed staff.

FR-4.5: The system shall maintain a complete audit trail of all manual changes made to automated estimates, including who made the change, what was changed, and when.

FR-4.6: Admins shall be able to filter and sort projects by status, modification type, date submitted, and estimated value.

FR-4.7: The system shall provide dashboard analytics on key operational metrics including automation rate, pricing accuracy, and processing times.

FR-4.8: Admins shall be able to update the pricing matrix (material costs, labor rates, markup tiers, grant eligibility criteria) through a configuration interface without requiring code changes. Each pricing matrix update shall create a new version with a full audit trail.

FR-4.9: The Client Advisory Team shall be able to request additional photos or information from clients through the system, which shall send an email notification to the client with clear instructions on what is needed.

FR-4.10: Admins shall be able to manually override the AI-generated modification type and scope before estimate generation.

FR-4.11: Admins shall be able to attach files (additional specifications, custom drawings, vendor quotes) to project records, with visibility controls for staff-only or client-visible files.

FR-4.12: The system shall maintain a canonical project status that maps to and displays BuilderTrend work order status and grant application status separately, with clear indication when external statuses differ from the internal InPlace AI status.

5. BuilderTrend Integration

FR-5.1: Upon refined estimate approval by the client, the system shall automatically create a work order in BuilderTrend with all relevant project details.

FR-5.2: The integration shall transfer client contact information, project scope, itemized pricing breakdown, and uploaded photos to BuilderTrend.

FR-5.3: The system shall handle BuilderTrend integration authentication securely and retry failed transfers automatically (up to 3 attempts with exponential backoff).

FR-5.4: If BuilderTrend supports status callbacks or webhooks, work order status updates from BuilderTrend shall be reflected back in the InPlace AI system. If BuilderTrend does not support automatic status synchronization, the system shall display a link to the BuilderTrend work order and the timestamp of the last manual status sync.

FR-5.5: The system shall generate a unique project identifier that links records across both InPlace AI and BuilderTrend platforms.

FR-5.6: If the BuilderTrend integration transfer fails after all retry attempts, the system shall generate a manual fallback export package (PDF quote + photos + intake summary + grant documents) for staff to manually upload into BuilderTrend, and shall log the failure with the following details: error summary, timestamp, affected project ID, number of attempts made, and a clear recommended next action for staff.

FR-5.7 (V2 - Post-MVP): Upon client acceptance of an refined estimate, the system shall create or update the corresponding BuilderTrend entity through to contract-ready stage, including at minimum: lead record, estimate record, and contract record linkage with all required fields populated for BuilderTrend's sales funnel automation.

6. Notification & Communication

FR-6.1: Clients shall receive email notifications at key stages (submission received with unique reference number, initial estimate range ready, refined estimate ready, consultation required to finalize budget, work order created, project scheduled).

FR-6.2: The Client Advisory Team shall receive notifications when projects require manual review.

FR-6.3: The system shall send an automated Admin Daily Digest email (configurable: enabled/disabled) that includes: (a) all new submissions received in the previous 24 hours, (b) all requests requiring staff action (e.g., manual review, BuilderTrend transfer failure, missing client information), and (c) any requests at risk of missing the refined estimate delivery time window (as defined in FR-2.5b).

FR-6.4: All email communications shall be accessible, mobile-friendly, and written in plain language suitable for seniors.

FR-6.5: The system shall maintain a complete communication history for each project, showing all emails sent, notifications delivered, and timestamps.

FR-6.6: Admins shall be able to send custom email messages to clients through the system, which shall be logged in the communication history.

FR-6.7: The system shall send automated reminder notifications to clients for pending administrative actions (such as providing additional information or documentation requested by staff) if no response is received within 7 days.

7. Client Dashboard & Request Management

FR-7.1: Clients shall have access to a personalized dashboard displaying all their submitted modification requests with current status.

FR-7.2: The dashboard shall display requests in chronological order with filtering options by status (pending, in review, initial estimate range ready, refined estimate ready, consultation required, in progress, completed).

FR-7.3: Clients shall be able to view detailed information for each request including submitted photos, modification type, submission date, and current status.

FR-7.4: Clients shall be able to download their estimates, grant documents, and final work orders directly from their dashboard through secure, time-limited download links.

FR-7.5: The dashboard shall provide a notification center showing recent updates and action items requiring client attention.

FR-7.6: Clients shall be able to view a complete history of all projects including completed and archived requests.

8. Estimate Review & Approval Workflow

FR-8.1: Upon client finalization of intake, the system shall display the initial estimate range in the client dashboard.

FR-8.2: Upon generation of the refined estimate (as defined in FR-2.5b), clients shall receive an email notification prompting them to review the detailed estimate in their dashboard.

FR-8.3: The estimate detail page shall clearly display the project scope, itemized pricing breakdown, tiered options (if applicable), timeline estimate, and grant eligibility information.

FR-8.4: Clients shall be able to formally accept or decline the **refined estimate** through the system interface.

FR-8.5: If a client declines the refined estimate, they shall be prompted to provide a reason (dropdown options: too expensive, timeline too long, scope not right, other with text field).

FR-8.6: Clients shall be able to request modifications or ask questions about an refined estimate, which shall create a notification for the Client Advisory Team.

FR-8.7: The system shall enforce that a work order can only be created in BuilderTrend after the client has formally accepted the refined estimate.

FR-8.8: If a client does not respond to a refined estimate within 14 days, the system shall send a reminder email and mark the estimate as "awaiting client response."

FR-8.9: A refined estimate not responded to within 30 days shall be automatically marked as "expired" with an option for clients to reactivate the request.

9. User Account Management

FR-9.1: Clients shall be able to reset their password through a secure email verification process.

FR-9.2: Clients shall be able to update their account profile information including name, email, phone number, and mailing address.

FR-9.3: Clients shall be able to request account deletion, which shall trigger a data retention workflow in compliance with PIPEDA (personal data deleted after 30 days unless active projects exist).

FR-9.4: Clients shall be able to export all their personal data in a standard format (PDF or CSV) upon request.

FR-9.5: The system shall support multiple authorized users (e.g., family members, caregivers) accessing the same client account with appropriate permission levels (view-only or full access).

FR-9.6: Clients shall be able to change their email address with verification of both old and new email addresses.

FR-9.7: The system shall enforce secure password requirements with clear guidance to clients during account creation and password changes.

10. Data Privacy & Retention

FR-10.1: The system shall support configurable retention policies for photos and personal data on declined or expired projects, with notification sent to clients prior to deletion. Default policy shall be 90 days with 14-day advance notice.

FR-10.2: For completed projects, the system shall archive photos and documents according to configurable retention policies before permanent deletion.

FR-10.3: The system shall maintain an audit log of all data access, modifications, and deletions for compliance purposes, including who accessed what data, when, and what action was performed.

FR-10.4: When a client requests account deletion, the system shall provide a grace period of 30 days during which the account can be reactivated before permanent deletion. Active projects shall prevent immediate deletion until resolved or transferred.

Non-Functional Requirements

1. Performance

NFR-1.1: The system shall maintain 99.5% uptime during core business hours (9:00 AM–5:00 PM EST, Monday–Friday), measured monthly.

NFR-1.2: The digital intake form shall reach an interactive/usable state within 3 seconds (p95) on standard broadband connections.

NFR-1.3: The system shall support at least 20 concurrent intake submissions, including concurrent photo uploads and background processing, without user-facing failures.

NFR-1.4: The AI estimation pipeline (photo analysis + scope inference + quote generation) shall complete within 60 minutes for 90% of submissions, and shall not prevent meeting the refined estimate delivery window defined in FR-2.5b.

NFR-1.5: The system shall support at least 100 submissions per day and a peak burst of 30 submissions per hour without performance degradation.

2. Security & Privacy

NFR-2.1: All data transmission shall be encrypted using TLS 1.3 or higher.

NFR-2.2: Personal data handling shall comply with applicable Canadian privacy legislation (including PIPEDA) and LandSeed policies.

NFR-2.3: The system shall enforce role-based access control (RBAC) across all features and all data (including file access).

NFR-2.4: Admin authentication shall support multi-factor authentication (MFA).

NFR-2.5: Client data retention and deletion shall be governed by configurable retention policies aligned to LandSeed compliance requirements, with secure deletion when retention expires.

NFR-2.6: The system shall log all access to sensitive client information for audit purposes.

NFR-2.7: Any third-party service used to process client photos/personal data shall require LandSeed approval and documented data-handling terms prior to production use.

NFR-2.8: The system shall disclose in plain language whether uploaded photos/personal data may be processed by third-party services and for what purpose.

NFR-2.9: The system shall support configurable data residency/storage-region settings aligned to LandSeed compliance requirements.

NFR-2.10: All stored client data and uploaded files shall be encrypted at rest.

NFR-2.11: Encryption keys shall be managed via a managed KMS, with access controls and a defined key rotation policy.

NFR-2.12: The system shall enforce rate limiting and abuse protection for login, password reset, form submission, and file upload endpoints.

NFR-2.13: Application logs shall avoid storing personal data and file contents; where identifiers are required for troubleshooting, they shall be masked/redacted or replaced with internal IDs.

3 Usability & Accessibility

NFR-3.1: The interface shall comply with WCAG 2.1 Level AA.

NFR-3.2: Text shall be resizable up to 200% without loss of functionality.

NFR-3.3: The intake form shall be fully usable with keyboard-only navigation.

NFR-3.4: Color contrast ratios shall meet accessibility standards for users with visual impairments.

NFR-3.5: The system shall use clear, jargon-free language suitable for non-technical users.

NFR-3.6: The interface shall be fully responsive and functional on mobile devices (phones and tablets).

4 Scalability

NFR-4.1: The architecture shall support horizontal scaling to handle increased load during peak periods.

NFR-4.2: The data model shall support growth to 10,000+ projects annually without requiring restructuring.

NFR-4.3: The AI/processing workload shall be deployable on scalable infrastructure to handle varying processing demand.

5. Maintainability

NFR-5.1: The codebase shall follow established coding standards and include sufficient documentation for maintenance and onboarding.

NFR-5.2: The system shall include automated testing with minimum 80% code coverage for core business logic.

NFR-5.3: Third-party dependencies shall be documented with version numbers and a defined update process.

NFR-5.4: The system shall provide structured error logging sufficient for troubleshooting and debugging.

6. Compatibility

NFR-6.1: The web application shall function correctly on the latest versions of Chrome, Firefox, Safari, and Edge.

NFR-6.2: The system shall support browsers up to 2 versions behind current release.

NFR-6.3: Mobile compatibility shall be ensured for iOS 14+ and Android 10+.

7. Reliability & Recovery

NFR-7.1: The system shall implement automated daily backups of all data with 30-day retention (configurable).

NFR-7.2: Backups shall be stored in geographically separate locations for disaster recovery.

NFR-7.3: The system shall have a Recovery Time Objective (RTO) of 4 hours (maximum time to restore service after an outage) and a Recovery Point Objective (RPO) of 1 hour (maximum acceptable data loss measured back from the time of failure).

NFR-7.4: Failed AI processing jobs shall be automatically retried up to 3 times before being flagged for manual review.

NFR-7.5: Backup restoration shall be tested at least monthly and results documented to validate RTO/RPO targets.

8. Observability & Notifications

NFR-8.1: The system shall provide monitoring and alerting for critical failures (AI job failures, BuilderTrend transfer failures, email delivery failures, file scanning failures).

NFR-8.2: Critical alerts shall notify designated admins/on-call within 15 minutes.

NFR-8.3: Outbound email shall be configured for deliverability (SPF/DKIM/DMARC) and include bounce handling to detect failed client notifications.

NFR-8.4: Audit logs shall be append-only and tamper-evident (i.e., log entries cannot be edited or deleted without detection), retained for a configurable period, and accessible only to authorized roles.